Theory

ISPhO

A2-1

English

Muon anomalous magnetic moment

Part A. Dynamics in magnetic field (4.5 points)

A1 (1.0pt)

(Formula) $\Delta v =$ (Value) $\Delta v =$

A2 (0.5pt)

(Formula) $E_k =$ (Value) $E_k =$

A3 (1.0pt)

(Formula) $R =$ (Value) $R =$

(Formula) $T =$ (Value) $T =$

A4 (0.8pt)

(Formula) $N =$ (Value) $N =$

A5 (0.7pt)

$d\vec{s}/dt =$

A6 (0.5pt)

$\omega_z =$

Part B. Thomas precession (3.5 points)

B1 (0.5pt)

$v_{Cx} =$ $v_{Cy} =$

B2 (0.7pt)

$v''_{x''} =$ $v''_{y''} =$

B3 (0.6pt)

B4 (1.0pt)

$\omega =$

B5 (0.7pt)

$\omega_{T,z} =$

Part C. Magnetic moment precession (2.0 points)

C1 (0.7pt)

$\omega_s =$

C2 (0.5pt)

$\omega_a =$

C3 (0.8pt)

(Formula) $a_\mu =$ (Value) $a_\mu =$